

When the Countable-Capture BCP Obstruction Exists

Run `fa_banach_001`, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Let X be a nonzero normed space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. We say that X has the *countable-capture property* if for every countable set $C \subset X$ there are a normed space Y_C , a contraction

$$E_C : X \rightarrow Y_C,$$

and $u_C \in S_X$ such that

$$E_C u_C = 0, \quad \|E_C c\| = \|c\| \quad (c \in C).$$

The previous packet

`2311.14257_countable_capture_bcp_obstruction`

proved that this property forces failure of BCP.

For $x \in X \setminus \{0\}$, write

$$J(x) = \{f \in X^* : \|f\| = 1, f(x) = \|x\|\}$$

for the norming functionals of x .

Statement

Theorem 1 (exact Banach-space characterization). *For a nonzero normed space X , the following are equivalent:*

1. X has the countable-capture property.
2. For every countable $C \subset X$, there exists $u \in S_X$ such that

$$\text{dist}(c, \mathbb{F}u) = \|c\| \quad (c \in C).$$

3. For every countable $C \subset X \setminus \{0\}$, there are functionals $f_c \in J(c)$, $c \in C$, such that

$$\bigcap_{c \in C} \ker f_c$$

contains a nonzero vector.

Theorem 2 (primitive-space source of countable capture). *Let A be a nonzero C^* -algebra. If $\text{Prim}(A)$ has no countable π -base, then the underlying Banach space of A has the countable-capture property.*

Corollary 1 (commutative exact classification). *Let K be a nonempty locally compact Hausdorff space. Then $C_0(K)$ has the countable-capture property if and only if K has no countable π -base.*

Corollary 2 (Hilbert spaces). *A nonzero Hilbert space H has the countable-capture property if and only if H is nonseparable.*

Corollary 3 (reduced group C^* -algebras). *For a discrete group G , $C_r^*(G)$ has the countable-capture property if and only if G is uncountable.*

Idea of the Proof

The contraction in the countable-capture definition is not mysterious. If a contraction kills u , it factors through the quotient $X/\mathbb{F}u$. Since the quotient map is the largest contractive map killing u , exact norm preservation of c is equivalent to

$$\|c + \mathbb{F}u\|_{X/\mathbb{F}u} = \|c\|,$$

that is, $\text{dist}(c, \mathbb{F}u) = \|c\|$. Hahn-Banach then says the same thing in dual language: each c has a norming functional vanishing on the same fresh direction u .

For C^* -algebras with no countable primitive π -base, the fresh direction comes from an open ideal. Given countably many centres a_n , form the countable family of primitive near-norm sets

$$\{P : \|a_n + P\| > \|a_n\| - \varepsilon\}.$$

Since these sets are not a π -base, there is a nonempty open set V containing none of them. A norm-one element u in the ideal for V vanishes in primitive quotients outside V , while every a_n is arbitrarily near normed in such quotients. Hence

$$\text{dist}(a_n, \mathbb{C}u) = \|a_n\|$$

for all n .

Proof of the Banach-Space Characterization

Proof of Theorem 1. Suppose first that X has countable capture, and fix a countable set $C \subset X$. Choose $E : X \rightarrow Y$ and $u \in S_X$ as in the definition. Let $q : X \rightarrow X/\mathbb{F}u$ be the quotient map. Since $E(u) = 0$, the map E factors through q : there is a contraction $\tilde{E} : X/\mathbb{F}u \rightarrow Y$ such that $E = \tilde{E}q$. Therefore, for every $c \in C$,

$$\|c\| = \|Ec\| \leq \|q(c)\| \leq \|c\|.$$

Thus $\|q(c)\| = \|c\|$, which is exactly $\text{dist}(c, \mathbb{F}u) = \|c\|$. This proves (1) implies (2).

Conversely, if (2) holds, take $Y = X/\mathbb{F}u$ and $E = q$, the quotient map. Then q is a contraction, $q(u) = 0$, and

$$\|q(c)\| = \text{dist}(c, \mathbb{F}u) = \|c\|$$

for every $c \in C$. Hence X has countable capture.

It remains to compare (2) and (3). Assume (2), and fix a countable $C \subset X \setminus \{0\}$. Choose $u \in S_X$ such that $\text{dist}(c, \mathbb{F}u) = \|c\|$ for all $c \in C$. For each c , the quotient element $q(c) \in X/\mathbb{F}u$ has norm $\|c\|$. By Hahn-Banach, choose a norm-one functional φ_c on $X/\mathbb{F}u$ such that

$$\varphi_c(q(c)) = \|q(c)\| = \|c\|.$$

Then $f_c = \varphi_c q$ belongs to $J(c)$, and $f_c(u) = 0$. Thus

$$u \in \bigcap_{c \in C} \ker f_c.$$

Conversely, assume (3). Given countable $C \subset X$, discard the zero vectors. Choose $f_c \in J(c)$ with a nonzero common kernel vector v , and put $u = v/\|v\|$. For $c \neq 0$ and every scalar λ ,

$$\|c - \lambda u\| \geq |f_c(c - \lambda u)| = \|c\|.$$

Taking the infimum over λ gives $\text{dist}(c, \mathbb{F}u) \geq \|c\|$, while the reverse inequality follows by taking $\lambda = 0$. Hence $\text{dist}(c, \mathbb{F}u) = \|c\|$ for all $c \in C$, so (2) holds. $\square$

Primitive-Ideal Capture

We use two standard facts. First, open subsets of $\text{Prim}(A)$ correspond to closed two-sided ideals: if $I \triangleleft A$, then

$$U_I = \{P \in \text{Prim}(A) : I \not\subseteq P\}$$

is open, and every open set arises this way. Second,

$$\|a\| = \sup_{P \in \text{Prim}(A)} \|a + P\|.$$

Lemma 1. *For $a \in A$ and $t > 0$, the set*

$$U(a, t) = \{P \in \text{Prim}(A) : \|a + P\| > t\}$$

is open in $\text{Prim}(A)$.

Proof. The element $b = (a^*a - t^2)_+$ satisfies

$$b + P = ((a + P)^*(a + P) - t^2)_+$$

in A/P . Hence $b \notin P$ if and only if $\|a + P\| > t$. Thus

$$U(a, t) = \{P : b \notin P\} = U_{\overline{AbA}},$$

which is open. □

Proof of Theorem 2. Let $C = \{a_1, a_2, \dots\} \subset A$ be countable. For every n with $r_n = \|a_n\| > 0$, and every integer m with $1/m < r_n$, set

$$U_{n,m} = \{P \in \text{Prim}(A) : \|a_n + P\| > r_n - 1/m\}.$$

By Lemma 1, these are nonempty open sets, and the whole family is countable. Since $\text{Prim}(A)$ has no countable π -base, there is a nonempty open set $V \subseteq \text{Prim}(A)$ such that no $U_{n,m}$ is contained in V .

Let $I \triangleleft A$ be the nonzero ideal with $V = U_I$. Choose $u \in I$ with $\|u\| = 1$. Fix n . If $a_n = 0$, then $\text{dist}(a_n, \mathbb{C}u) = 0 = \|a_n\|$. If $r_n > 0$, then for all sufficiently large m we may choose

$$P_{n,m} \in U_{n,m} \setminus V.$$

Because $P_{n,m} \notin V = U_I$, we have $I \subseteq P_{n,m}$, hence $u \in P_{n,m}$. Therefore for every scalar λ ,

$$\|a_n - \lambda u\| \geq \|a_n - \lambda u + P_{n,m}\| = \|a_n + P_{n,m}\| > r_n - \frac{1}{m}.$$

Letting $m \rightarrow \infty$ gives

$$\|a_n - \lambda u\| \geq r_n \quad (\lambda \in \mathbb{C}).$$

Thus $\text{dist}(a_n, \mathbb{C}u) \geq r_n$, and equality follows by taking $\lambda = 0$. Theorem 1 now gives countable capture. □

Model Classifications

Proof of Corollary 1. For $A = C_0(K)$, the primitive ideal space is homeomorphic to K . If K has no countable π -base, Theorem 2 gives countable capture.

Conversely, if K has a countable π -base, then $C_0(K)$ has UBCP by the commutative $C_0(K)$ theorem of Liu–Liu–Lu–Zheng. Since countable capture implies failure of BCP, $C_0(K)$ cannot have countable capture. □

Proof of Corollary 2. If H is nonseparable and $C \subset H$ is countable, the closed linear span $L = \overline{\text{span}}C$ is separable and hence proper. Choose $u \in S_H \cap L^\perp$. Then for every $c \in C$,

$$\text{dist}(c, \mathbb{F}u) = \|c\|.$$

So H has countable capture by Theorem 1.

If H is separable, then H has UBCP because every separable normed space has UBCP: choose a countable dense subset (d_j) of S_H and use the balls $B(2d_j, 3/2)$. Since countable capture forces failure of BCP, separable H cannot have countable capture. □

Proof of Corollary 3. If G is countable, then $C_r^*(G)$ is separable, hence has UBCP, so it does not have countable capture.

If G is uncountable and $C \subset C_r^*(G)$ is countable, the reduced group packet shows that $C \subset C_r^*(H)$ for some countable subgroup $H \leq G$. Choose $g \in G \setminus H$. The canonical subgroup expectation

$$E_H : C_r^*(G) \rightarrow C_r^*(H)$$

fixes $C_r^*(H)$ and kills λ_g . Thus E_H and $u = \lambda_g$ witness countable capture for C . $\square$

Scope and Remaining Wall

This packet classifies the countable-capture mechanism itself in Banach-space terms and identifies broad exact model classes. It also upgrades the primitive π -base obstruction: if the primitive space has no countable π -base, then the obstruction is literally countable capture.

It does not classify all C^* -algebras with countable capture. The hard remaining region is where $\text{Prim}(A)$ already has a countable π -base. This includes simple nonseparable algebras, atomless von Neumann factors, and corona/reduced-product examples. In that region the obstruction must come from an internal norm-visibility mechanism, such as subgroup expectations, annular diagonal maps, or von Neumann corner decompositions, rather than from the topology of $\text{Prim}(A)$.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, countable capture, quotient by a line, Birkhoff–James orthogonality, primitive ideal space, π -base, conditional expectation, and reduced group C^* -algebras. Web searches on June 17, 2026 for countable capture, Birkhoff–James orthogonality with BCP, primitive ideal spaces with BCP, and $C(K)$ π -base BCP found the source paper and the commutative $C_0(K)$ classification paper, but no exact statement of the present characterization.

Human Review Focus

Reviewers should check:

1. the quotient factorization in Theorem 1;
2. the Hahn–Banach equivalence with common-kernel norming functionals;
3. the primitive-space argument, especially the choice of V and the quotient estimates outside V ;
4. the use of the known $C_0(K)$ BCP theorem in the commutative converse.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.

- [3] Run `fa.banach_001`, lane 19, *A Countable-Capture Obstruction for the Ball-Covering Property*, solution packet `2311.14257_countable_capture_bcp_obstruction`, 2026.
- [4] Run `fa.banach_001`, lane 19, *A Primitive-Ideal Π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
- [5] Run `fa.banach_001`, lane 19, *Reduced Group C^* -Algebras and the Ball-Covering Property*, solution packet `2311.14257_reduced_group_cstar_bcp_countable`, 2026.